Binomial coefficients mod p: Kummer, Lucas and valuations

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L5-X6 Level 5 Advanced · Number theory · Beyond the textbook

Binomial coefficients and primes digit by digit: Legendre's formula as a digit sum, Kummer's carries, Lucas's theorem, the prime factors of central binomial coefficients, and Wolstenholme's congruences modulo p^2 and p^3.
By the end you can compute the exact power of a prime dividing a binomial coefficient by counting carries, and count coefficients with a given valuation; evaluate binomial coefficients modulo a prime with Lucas's theorem and use it in proofs; use Wolstenholme's theorem and the central binomial facts with correct hypotheses.
Prerequisites: Orders, Fermat and Euler (L4-NT1), Quadratic residues (L5-NT1). **Contests:** BMO Round 2, EGMO and IMO selection, USAMO, IMO. **Time:** about 180 minutes.

1 Lesson

How many of the binomial coefficients $\binom{2026}{k}$, $0 \le k \le 2026$, are odd? How many are not divisible by 3?

In binary $2026 = 11111101010_2$, and in base 3 it is 2210001_3. The answers are $2^8 = 256$ and $3 \cdot 3 \cdot 2 \cdot 2 = 36$: one more than each binary digit, multiplied together, and the same in base 3. This unit explains why. The tools are four classical results about binomial coefficients and primes: Legendre's formula (1808) in its digit-sum form, Kummer's theorem (1852) that the power of p dividing $\binom{m+n}{m}$ is the number of carries when m and n are added in base p, Lucas's theorem (1878) which computes $\binom{n}{k}$ modulo p digit by digit, and Wolstenholme's theorem (1862) which goes from modulo p to modulo p^3. The standard approach treats $\binom{n}{k}$ as a quotient of factorials and computes one valuation at a time with Legendre's formula. That works for one coefficient; it gives no hold on questions about all k at once, or on residues rather than divisibility. We assume L4-NT1 (Legendre's formula, Fermat) and use nothing from L5-NT1 beyond working with inverses modulo p.

Notation: p is a prime, $v_p(n)$ is the exponent of p in n, and $s_p(n)$ is the sum of the base-p digits of n. When $n = \sum n_i p^i$ we call n_i the i-th digit of n.

1.1 Legendre's formula and Kummer's theorem

Theorem 1 (Legendre).
$v_p(n!) = \sum_{i \ge 1} \left\lfloor \frac{n}{p^i} \right\rfloor = \frac{n - s_p(n)}{p-1}$.

Proof. The first form is L4-NT1. For the second, write $n = \sum_j n_j p^j$. Then $\lfloor n/p^i \rfloor = \sum_{j \ge i} n_j p^{j-i}$, and summing over $i \ge 1$ the digit n_j appears with weight $p^{j-1} + \cdots + p + 1 = \frac{p^j - 1}{p-1}$. So the sum is $\sum_j n_j \frac{p^j - 1}{p-1} = \frac{n - s_p(n)}{p-1}$. □

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Theorem 2 (Kummer).
For non-negative integers m and n, $v_p\binom{m+n}{m}$ is the number of carries when m and n are added in base p.

Proof. Let a_i, b_i, r_i be the digits of $m, n, m+n$, and let $c_i \in \{0,1\}$ be the carry into position i, with $c_0 = 0$. The addition algorithm says $a_i + b_i + c_i = r_i + pc_{i+1}$ for every i. Sum over all positions, taking enough of them that the last carry is 0. Writing $C = \sum_i c_i = \sum_i c_{i+1}$ for the number of carries, we get $s_p(m) + s_p(n) + C = s_p(m+n) + pC$, that is $s_p(m) + s_p(n) - s_p(m+n) = (p-1)C$. By Theorem 1,

$$v_p\binom{m+n}{m} = \frac{(m+n-s_p(m+n)) - (m-s_p(m)) - (n-s_p(n))}{p-1} = \frac{s_p(m)+s_p(n)-s_p(m+n)}{p-1} = C. \qquad \square$$

For $\binom{n}{k}$ the two summands are k and $n-k$. For $p=2$ and $\binom{2n}{n}$ every 1 in n produces a carry, so $v_2\binom{2n}{n} = s_2(n)$. For odd p the incoming carry matters: with $p=5$ and $n = 13 = 23_5$, the units give $3+3=6$, a carry, and then $2+2+1=5$ carries again, so $v_5\binom{26}{13} = 2$ although the digit 2 is less than $\frac{5}{2}$.

Corollary 3.
If $p^e \mid \binom{n}{k}$ then $p^e \le n$. Consequently, for $2 \le k \le n-2$ the number $\binom{n}{k}$ is never a prime power.

Proof. Let $p^L \le n < p^{L+1}$. A carry out of position L when adding k and $n-k$ would make $n \ge p^{L+1}$, so carries occur only out of positions $0, \dots, L-1$, and $e \le v_p\binom{n}{k} \le L$. For the second part, if $\binom{n}{k} = p^e$ with $2 \le k \le n-2$, then $n \ge 4$ and $p^e = \binom{n}{k} \ge \binom{n}{2} = \frac{n(n-1)}{2} > n$, a contradiction. $\square$

The same bound follows from Legendre's formula, since each term $\lfloor \frac{n}{p^i} \rfloor - \lfloor \frac{k}{p^i} \rfloor - \lfloor \frac{n-k}{p^i} \rfloor$ is 0 or 1 and vanishes once $p^i > n$; Theorem 5 below uses it in this form.

Example 1.
For how many k with $0 \le k \le 2026$ is $\binom{2026}{k} \equiv 2 \pmod 4$?

Solution.
Computing v_2 of each coefficient with Legendre's formula is out of the question. By Kummer we need exactly one carry when k and $l = 2026 - k$ are added in binary. Use the carry equation $k_i + l_i + c_i = n_i + 2c_{i+1}$ with $n = 2026$. Suppose the only carry is out of position i. Then $c_i = 0$ and $c_{i+1} = 1$, so $k_i + l_i = n_i + 2$, which forces $n_i = 0$ and $k_i = l_i = 1$. At position $i+1$ there is no carry out, so $k_{i+1} + l_{i+1} + 1 = n_{i+1}$, which forces $n_{i+1} = 1$ and $k_{i+1} = l_{i+1} = 0$. At every other position $k_j + l_j = n_j$, so $k_j = 0$ when $n_j = 0$, and k_j is free when $n_j = 1$. Conversely each such choice of k gives exactly one carry.
So the count is (number of positions i with $n_i = 0$ and $n_{i+1} = 1$) times $2^{s_2(n)-1}$. Here $2026 = 11111101010_2$ has ones at positions $10, 9, 8, 7, 6, 5, 3, 1$ and zeros at $4, 2, 0$, and each zero has a 1 directly above it. So the answer is $3 \cdot 2^7 = 384$.
Remark. The same method counts k with $v_2\binom{n}{k} = j$ for any small j: describe the possible carry patterns and count the free digits.

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1.2 Lucas's theorem

Theorem 4 (Lucas).
If $n = \sum n_i p^i$ and $k = \sum k_i p^i$ with digits $0 \le n_i, k_i \le p-1$, then

$$\binom{n}{k} \equiv \prod_i \binom{n_i}{k_i} \pmod{p},$$

where $\binom{a}{b} = 0$ when $b > a$. In particular $p \nmid \binom{n}{k}$ if and only if $k_i \le n_i$ for every i, and the number of such k is $\prod_i (n_i + 1)$.

Proof. Since $p \mid \binom{p}{j}$ for $0 < j < p$, we have $(1+x)^p \equiv 1 + x^p$, and by induction $(1+x)^{p^i} \equiv 1 + x^{p^i}$, as polynomials with coefficients modulo p. Hence

$$(1+x)^n = \prod_i \left((1+x)^{p^i}\right)^{n_i} \equiv \prod_i \left(1+x^{p^i}\right)^{n_i} = \prod_i \sum_{j=0}^{n_i} \binom{n_i}{j} x^{jp^i}.$$

Expanding the product, choosing $j = j_i$ from the i-th factor gives the monomial $x^{\sum j_i p^i}$, and since $0 \le j_i \le n_i \le p-1$, base-$p$ representation is unique, so x^k arises only from $j_i = k_i$ for all i, and not at all if some $k_i > n_i$. Comparing coefficients of x^k gives the theorem. □

Lucas gives the residue; Kummer gives the exact power. They agree on divisibility: there are no carries in $k + (n-k)$ exactly when every digit of k is at most the corresponding digit of n. Two consequences to remember: $\binom{pn}{pk} \equiv \binom{n}{k} \pmod{p}$, and $\binom{p-1}{k} \equiv (-1)^k \pmod{p}$, the latter because $\binom{p-1}{k} = \prod_{j=1}^{k} \frac{p-j}{j}$.

Example 2.
Let p be an odd prime. Find all $n \ge 0$ such that $\binom{n}{k} \equiv (-1)^k \pmod{p}$ for every $0 \le k \le n$.

Solution.
The condition says in particular that no $\binom{n}{k}$ is divisible by p, and Lucas turns that into a condition on the digits of n; then one well-chosen k decides the top digit. Small cases first: $n = p-1$ works by the remark above, and $n = p^2 - 1$ works by Lucas with both digits $p-1$. Try $n = p^m - 1$ in general: all its m digits are $p-1$, so $\binom{n}{k} \equiv \prod_i \binom{p-1}{k_i} \equiv (-1)^{s_p(k)}$, and $s_p(k) \equiv k \pmod 2$ because p^i is odd. So $n = p^m - 1$ works for every $m \ge 0$.
Conversely, suppose n works and let n have top digit $n_L \ge 1$ in position L. No $\binom{n}{k}$ is divisible by p. If some digit n_i with $i < L$ were less than $p-1$, then $k = (p-1)p^i$ satisfies $k < p^{i+1} \le n$ and has digit $p-1 > n_i$ in position i, so $p \mid \binom{n}{k}$ by Theorem 4, a contradiction. So $n = n_L p^L + (p^L - 1)$. Now take $k = p^L \le n$: by Lucas $\binom{n}{p^L} \equiv \binom{n_L}{1} = n_L$, while $(-1)^{p^L} = -1$, so $n_L \equiv -1 \pmod{p}$, that is $n_L = p-1$ and $n = p^{L+1} - 1$. Together with $n = 0 = p^0 - 1$, the answer is $n = p^m - 1$, $m \ge 0$.
Remark. The middle step is worth remembering on its own: no $\binom{n}{k}$ is divisible by p if and only if $n = ap^m - 1$ with $1 \le a \le p$.

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1.3 Central binomial coefficients

Theorem 5.
Let $n \geq 1$ and let p be prime. Then $v_p\binom{2n}{n} = \sum_{i\geq 1}\left(\left\lfloor \frac{2n}{p^i}\right\rfloor - 2\left\lfloor \frac{n}{p^i}\right\rfloor\right)$, each term is 0 or 1, and the terms vanish once $p^i > 2n$. Consequently:

1. $p^{v_p\binom{2n}{n}} \leq 2n$, so $\binom{2n}{n}$ divides $\operatorname{lcm}(1,2,\ldots,2n)$, and $v_p\binom{2n}{n} \leq 1$ when $p > \sqrt{2n}$;
2. every prime p with $n < p \leq 2n$ divides $\binom{2n}{n}$ exactly once;
3. if $n \geq 3$, no prime p with $\frac{2n}{3} < p \leq n$ divides $\binom{2n}{n}$.

Proof. The formula is Theorem 1 applied to $(2n)!/(n!)^2$, and $\lfloor 2x\rfloor - 2\lfloor x\rfloor \in \{0,1\}$ for real x. Part 1 follows because only the indices with $p^i \leq 2n$ contribute. For part 2, $p^2 \geq 2p > 2n$, so only $i = 1$ counts, and $\lfloor \frac{2n}{p}\rfloor - 2\lfloor \frac{n}{p}\rfloor = 1 - 0$. For part 3, $2 \leq \frac{2n}{p} < 3$ gives $\lfloor \frac{2n}{p}\rfloor - 2\lfloor \frac{n}{p}\rfloor = 2 - 2 = 0$, and $p^2 > 2n$ (from $p > \frac{2n}{3}$ when $n \geq 5$, and by inspection, $p = 3$, when $n = 3, 4$). □

As an illustration, take $n = 30$. The primes $31, 37, 41, 43, 47, 53, 59$ in $(30, 60]$ divide $\binom{60}{30}$ once, and 23 and 29, in $(20, 30]$, do not divide it. For the small primes count carries in $30 + 30$: in base 2, $30 = 11110_2$ gives $v_2 = s_2(30) = 4$; in bases 3 and 5, $30 = 1010_3 = 110_5$ gives no carries; in base 7, $30 = 42_7$ gives one carry; in bases 11, 13, 17 and 19, 30 is 28_{11}, 24_{13}, $(1, 13)_{17}$ and $(1, 11)_{19}$, giving one, none, one and one carry. Hence

$$\binom{60}{30} = 2^4 \cdot 7 \cdot 11 \cdot 17 \cdot 19 \cdot 31 \cdot 37 \cdot 41 \cdot 43 \cdot 47 \cdot 53 \cdot 59,$$

found without computing the number itself.

These are the facts behind Erdős's proof of Bertrand's postulate, which combines them with $\binom{2n}{n} \geq \frac{4^n}{2n+1}$. They also limit what can be proved: whether there are infinitely many n with $\binom{2n}{n}$ coprime to 105 is an open problem of Erdős and Graham, so a contest problem of that shape must have extra structure.

1.4 Wolstenholme's theorem

For rationals r, s whose denominators are not divisible by p, we write $r \equiv s \pmod{p^e}$ when the numerator of $r - s$ in lowest terms is divisible by p^e. These congruences can be added, multiplied, and divided by integers prime to p. Write $H = \sum_{k=1}^{p-1} \frac{1}{k}$.

Theorem 6 (Wolstenholme).
Let $p \geq 5$ be prime. Then (a) $\sum_{k=1}^{p-1} \frac{1}{k^2} \equiv 0 \pmod{p}$; (b) $H \equiv 0 \pmod{p^2}$; (c) $\binom{2p-1}{p-1} \equiv 1 \pmod{p^3}$, equivalently $\binom{2p}{p} \equiv 2 \pmod{p^3}$.

Proof. (a) As k runs over $1, \ldots, p-1$, so does the residue of $\frac{1}{k}$ modulo p, so the sum is congruent to $\sum k^2 = \frac{(p-1)p(2p-1)}{6} \equiv 0 \pmod{p}$, using $p \geq 5$ to divide by 6. (b) Pair k with $p - k$: $2H = \sum_k \left(\frac{1}{k} + \frac{1}{p-k}\right) = p\sum_k \frac{1}{k(p-k)}$, and $\frac{1}{k(p-k)} \equiv -\frac{1}{k^2} \pmod{p}$, so by (a) the last sum is $\equiv 0 \pmod{p}$ and $2H \equiv 0 \pmod{p^2}$. (c) We have

$$\binom{2p-1}{p-1} = \prod_{k=1}^{p-1} \frac{p+k}{k} = \prod_{k=1}^{p-1}\left(1 + \frac{p}{k}\right) = 1 + pe_1 + p^2e_2 + p^3(\cdots),$$

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where $e_1 = H$, $e_2 = \sum_{i<j} \frac{1}{ij} = \frac{H^2 - \sum 1/k^2}{2}$, and the omitted part has denominator prime to p. By (b), $pe_1 \equiv 0 \pmod{p^3}$, and by (a) and (b), $e_2 \equiv 0 \pmod{p}$. □

State it with care. It needs $p \geq 5$: $\binom{5}{2} = 10 \not\equiv 1 \pmod{27}$ and $\binom{3}{1} = 3 \not\equiv 1 \pmod{8}$ (Babbage's 1819 congruence modulo p^2 holds for all $p \geq 3$). Whether a composite n can satisfy $\binom{2n-1}{n-1} \equiv 1 \pmod{n^3}$ is open. The primes with the stronger congruence modulo p^4 are called Wolstenholme primes; only 16843 and 2124679 are known.

Example 3.
Let $p \geq 5$ be prime and a a positive integer. Prove that $\binom{ap}{p} \equiv a \pmod{p^3}$, and find the remainder when $\binom{1001}{7}$ is divided by 343.

Solution.
Lucas only gives $\binom{ap}{p} \equiv \binom{a}{1} = a \pmod{p}$. To get further, write the coefficient as a product as in Theorem 6:

$$\binom{ap}{p} = \prod_{k=1}^{p} \frac{(a-1)p+k}{k} = a \prod_{k=1}^{p-1} \left(1 + \frac{(a-1)p}{k}\right),$$

where the factor $k = p$ is $\frac{ap}{p} = a$. Expanding, the product is $1 + (a-1)pe_1 + (a-1)^2p^2e_2 + p^3(\cdots)$ with the same e_1, e_2 as before, so it is $\equiv 1 \pmod{p^3}$ by Theorem 6. Hence $\binom{ap}{p} \equiv a \pmod{p^3}$. Since $1001 = 143 \cdot 7$, the remainder of $\binom{1001}{7}$ modulo $7^3 = 343$ is 143.
Remark. More generally $\binom{ap}{bp} \equiv \binom{a}{b} \pmod{p^3}$ for $p \geq 5$ (Ljunggren, 1952); the proof needs more bookkeeping, and we do not use it.

Common traps.

- Kummer counts carries in $k + (n-k)$, not in $k + n$, and the carry into a position counts: in base 5, $13 + 13$ has two carries.
- Lucas's theorem is for a prime modulus only. Modulo 4 the digit rule fails: $\binom{4}{2} = 6 \equiv 2$, while the base-2 digit product is 0.
- Wolstenholme needs $p \geq 5$. Say which congruence you mean when fractions appear, and check that every denominator is prime to p.
- Legendre's digit-sum form divides by $p - 1$; for $p = 2$ it reads $v_2(n!) = n - s_2(n)$.
- A marker wants the base-p expansions written out and the carry positions identified, not just a final count.

1.5 How to spot this in a contest

1. A prime power dividing a binomial coefficient, a product of consecutive integers, or a quotient of factorials: Kummer or Legendre.
2. A binomial coefficient with huge entries modulo a small prime: Lucas.
3. Counting entries of Pascal's triangle with a divisibility property: digit counts from Lucas and Kummer.
4. Harmonic sums, $\binom{2p}{p}$ or $\binom{ap}{bp}$ modulo p^2 or p^3: Wolstenholme's pairing $k \leftrightarrow p - k$.
5. $\binom{2n}{n}$, $\operatorname{lcm}(1, \ldots, n)$, or primes in $(n, 2n]$: Theorem 5.

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6. Showing that a quotient of factorials is an integer: compare valuations prime by prime, which reduces to an inequality between floor functions.

2 Practice problems

The problems are roughly in order of difficulty, from ⋆⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆⋆ Find the smallest positive integer n such that 2026^2 divides $\binom{2n}{n}$. (Note that 1013 is prime.)

Problem 2 ⋆⋆ Let p be prime and $n \geq 1$. Prove that $v_p\left(\binom{p^n}{k}\right) = n - v_p(k)$ for every $1 \leq k \leq p^n$.

Problem 3 ⋆⋆ Let p be prime and let n have base-p digits $n_0, n_1, \ldots, n_L$. Prove that

$$\sum_{k=0}^{n} \binom{n}{k}^{p-1} \equiv \prod_{i=0}^{L} (n_i + 1) \pmod{p}.$$

Deduce that if every base-3 digit of n is 0 or 1, then $\binom{2n}{n} \equiv (-1)^t \pmod 3$, where t is the number of digits equal to 1, and that otherwise $3 \mid \binom{2n}{n}$.

Problem 4 ⋆⋆⋆ For a positive integer m let $L_m = \operatorname{lcm}(1, 2, \ldots, m)$. Prove that $\frac{L_{2n}}{L_n}$ divides $\binom{2n}{n}$ for every $n \geq 1$.

Problem 5 ⋆⋆⋆ For a prime p, write $\sum_{k=1}^{p-1} \frac{1}{k^3} = \frac{a}{b}$ in lowest terms. Find all primes p for which p^2 divides a.

Problem 6 ⋆⋆⋆⋆ Let p be a prime and n a positive integer, and write $n + 1 = cp^m$ with $m \geq 0$ and $p \nmid c$. Prove that none of $\binom{n}{0}, \binom{n}{1}, \ldots, \binom{n}{n}$ is divisible by p^2 if and only if $c < p^2$.

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **AIME II 2006 Problem 3** ⋆
   Write the product as a quotient of factorials and apply Legendre's formula twice.
2. **AIME 1983 Problem 8** ⋆
   For a candidate prime, compare its exponent in the numerator with twice its exponent in each factorial of the denominator.
3. **HMMT February 2015 Guts Round Problem 27** ⋆⋆
   The range is chosen to be 0 to 3^4 - 1. Write a and b in base 3.
4. **IMO 1972 Problem 3** ⋆⋆
   Compare the exponent of each prime on top and bottom; this reduces to an inequality between floor functions.
5. **APMO 2006 Problem 3** ⋆⋆⋆
   Express the count with a single binomial coefficient, then expand it as a product and use Wolstenholme's theorem.

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6. **IMO 2019 Problem 4** ⋆⋆⋆
   Compare the power of 2 on both sides using Legendre's formula, then compare sizes.

7. **IMO 1979 Problem 1** ⋆⋆⋆
   Turn the alternating sum into a sum of reciprocals over a range, then pair terms as in the proof of Wolstenholme's theorem.

8. **HMMT February 2022 Algebra and Number Theory Problem 10** ⋆⋆⋆⋆
   Use the digit-sum form of Legendre's formula to read the parity of the exponent from the base-p digits.

9. **USAMO 2010 Problem 5** ⋆⋆⋆⋆
   Use partial fractions, then pair terms whose denominators add up to a multiple of p.

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4 Hints

1. $2026^2 = 4 \cdot 1013^2$. Count carries in $n + n$ in base 1013, remembering the incoming carry.
2. Use $k\binom{p^n}{k} = p^n\binom{p^n-1}{k-1}$, and look at the digits of $p^n - 1$.
3. Fermat's little theorem turns each term into 0 or 1. For the deduction, $\sum_k \binom{n}{k}^2 = \binom{2n}{n}$.
4. $v_p(L_m)$ is the largest e with $p^e \le m$. When does $v_p(L_{2n})$ exceed $v_p(L_n)$, and which term of Theorem 5's sum is then equal to 1?
5. Pair k with $p - k$ as in the proof of Wolstenholme's theorem, and reduce to a sum of fourth powers modulo p. Small primes need separate treatment.
6. By Kummer you need the largest number of carries in $k + (n - k)$ over all k. Show that no carry can start at a position where n has digit $p - 1$, and that once a carry starts it can be kept going up to the top digit.

5 Solutions

Problem 1

Find the smallest positive integer n such that 2026^2 divides $\binom{2n}{n}$. (Note that 1013 is prime.)

We have $2026^2 = 2^2 \cdot 1013^2$. By Kummer, $v_2\binom{2n}{n} = s_2(n)$ and $v_{1013}\binom{2n}{n}$ is the number of carries in $n + n$ in base 1013. Write $n = a_1 \cdot 1013 + a_0$ when $n < 1013^2$. A carry out of position 0 happens if and only if $2a_0 \ge 1013$, that is $a_0 \ge 507$. A carry out of position 1 happens if and only if $2a_1 + c_1 \ge 1013$, where c_1 is the carry from position 0; this needs $a_1 \ge 507$, or $a_1 = 506$ and $c_1 = 1$.

If $n < 1013^2$, two carries need both of these, so $a_0 \ge 507$ and $a_1 \ge 506$, giving $n \ge 506 \cdot 1013 + 507 = 513085$. If $n \ge 1013^2$ then $n > 513085$. For $n = 513085$ there are two carries, so $1013^2 \mid \binom{2n}{n}$; and n is odd and greater than 1, so $s_2(n) \ge 2$ and $4 \mid \binom{2n}{n}$. Hence the answer is 513085.

Answer: 513085

Problem 2

Let p be prime and $n \ge 1$. Prove that $v_p\binom{p^n}{k} = n - v_p(k)$ for every $1 \le k \le p^n$.

From $\binom{N}{k} = \frac{N}{k}\binom{N-1}{k-1}$ with $N = p^n$ we get $k\binom{p^n}{k} = p^n\binom{p^n-1}{k-1}$. In base p, the number $p^n - 1$ has n digits, all equal to $p - 1$, and $0 \le k - 1 < p^n$ has at most n digits. So every digit of $k - 1$ is at most the corresponding digit of $p^n - 1$, and by Lucas's theorem $\binom{p^n-1}{k-1} \equiv \prod_i \binom{p-1}{j_i} \not\equiv 0 \pmod p$, where j_i are the digits of $k - 1$ (each factor is ± 1 modulo p). Taking v_p of both sides, $v_p(k) + v_p\binom{p^n}{k} = n + 0$, which is the claim.

(Kummer gives the same: if $v_p(k) = a$, adding k and $p^n - k$ in base p produces a carry out of each of the positions $a, a + 1, \ldots, n - 1$ and nowhere else.)

Problem 3

Let p be prime and let n have base-p digits $n_0, n_1, \ldots, n_L$. Prove that

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$$\sum_{k=0}^{n} \binom{n}{k}^{p-1} \equiv \prod_{i=0}^{L} (n_i + 1) \pmod{p}.$$

Deduce that if every base-3 digit of n is 0 or 1, then $\binom{2n}{n} \equiv (-1)^t \pmod 3$, where t is the number of digits equal to 1, and that otherwise $3 \mid \binom{2n}{n}$.

By Fermat's little theorem, $\binom{n}{k}^{p-1} \equiv 1 \pmod p$ if $p \nmid \binom{n}{k}$, and $\equiv 0$ otherwise. So the sum is congruent to the number of k in $[0, n]$ with $p \nmid \binom{n}{k}$, which by Lucas's theorem is $\prod_i (n_i + 1)$.

For $p = 3$, the identity $\sum_{k=0}^{n} \binom{n}{k}\binom{n}{n-k} = \binom{2n}{n}$ (choose n objects from two groups of n) gives $\binom{2n}{n} = \sum_k \binom{n}{k}^2 \equiv \prod_i (n_i + 1) \pmod 3$. If some digit n_i equals 2, the product contains the factor 3, so $3 \mid \binom{2n}{n}$. If every digit is 0 or 1, the product is $2^t \equiv (-1)^t \pmod 3$. (The first case agrees with Kummer: a digit 2 gives $2 + 2 \geq 3$, a carry.)

Problem 4

For a positive integer m let $L_m = \operatorname{lcm}(1, 2, \ldots, m)$. Prove that $\frac{L_{2n}}{L_n}$ divides $\binom{2n}{n}$ for every $n \geq 1$.

For a prime p, $v_p(L_m)$ is the largest e with $p^e \leq m$. Fix p and let e be the largest exponent with $p^e \leq n$, so $n < p^{e+1}$. Then $p^{e+2} \geq 2p^{e+1} > 2n$, so $v_p(L_{2n})$ is e or $e + 1$, and $v_p(L_{2n}/L_n) \in \{0, 1\}$. It is 1 exactly when $n < p^{e+1} \leq 2n$.

In that case consider the term $i = e + 1$ in Theorem 5: $\left\lfloor \frac{2n}{p^{e+1}} \right\rfloor - 2\left\lfloor \frac{n}{p^{e+1}} \right\rfloor = 1 - 0 = 1$, because $1 \leq \frac{2n}{p^{e+1}} < 2$ and $\frac{n}{p^{e+1}} < 1$. All the other terms are 0 or 1, so $v_p\binom{2n}{n} \geq 1 = v_p(L_{2n}/L_n)$. When $v_p(L_{2n}/L_n) = 0$ there is nothing to prove. As this holds for every prime p, $\frac{L_{2n}}{L_n}$ divides $\binom{2n}{n}$. Together with Theorem 5 this gives $\frac{L_{2n}}{L_n} \mid \binom{2n}{n} \mid L_{2n}$.

Problem 5

For a prime p, write $\sum_{k=1}^{p-1} \frac{1}{k^3} = \frac{a}{b}$ in lowest terms. Find all primes p for which p^2 divides a.

Let $S = \sum_{k=1}^{p-1} \frac{1}{k^3}$. Its denominator divides a product of numbers less than p, so $p^2 \mid a$ if and only if $S \equiv 0 \pmod{p^2}$ in the sense of the lesson. Small cases: for $p = 2$, $S = 1$; for $p = 3$, $S = 1 + \frac{1}{8} = \frac{9}{8}$, and $9 \mid 9$; for $p = 5$, $S = 1 + \frac{1}{8} + \frac{1}{27} + \frac{1}{64} = \frac{2035}{1728}$ and $2035 = 5 \cdot 11 \cdot 37$, so $25 \nmid a$.

Now let $p \geq 5$. For $1 \leq k \leq p - 1$,

$$\frac{1}{k^3} + \frac{1}{(p-k)^3} = \frac{(p-k)^3 + k^3}{k^3(p-k)^3} = \frac{p\,(p^2 - 3pk + 3k^2)}{k^3(p-k)^3}.$$

Since the numerator has a factor p, only the rest of the fraction modulo p matters modulo p^2, and modulo p we have $p^2 - 3pk + 3k^2 \equiv 3k^2$ and $(p-k)^3 \equiv -k^3$. So the pair contributes $\equiv -\frac{3p}{k^4} \pmod{p^2}$, and summing over all k (each pair is counted twice),

$$2S \equiv -3p \sum_{k=1}^{p-1} \frac{1}{k^4} \pmod{p^2}.$$

As k runs over $1, \ldots, p-1$ so does the residue of $\frac{1}{k}$, so $\sum \frac{1}{k^4} \equiv \sum k^4 \pmod p$, and

$$\sum_{k=1}^{p-1} k^4 = \frac{(p-1)p(2p-1)(3p^2 - 3p - 1)}{30}.$$

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For $p \geq 7$ the prime p does not divide 30, so this is divisible by p, hence $2S \equiv 0 \pmod{p^2}$ and $S \equiv 0 \pmod{p^2}$. (For $p = 5$ the formula gives $354 \equiv 4 \pmod{5}$, so $2S \equiv -12 \cdot 5 \not\equiv 0 \pmod{25}$, which agrees with the direct computation.) Hence $p^2 \mid a$ exactly for $p = 3$ and all primes $p \geq 7$.

Answer: $p = 3$ and every prime $p \geq 7$

Problem 6

Let p be a prime and n a positive integer, and write $n + 1 = cp^m$ with $m \geq 0$ and $p \nmid c$. Prove that none of $\binom{n}{0}, \binom{n}{1}, \ldots, \binom{n}{n}$ is divisible by p^2 if and only if $c < p^2$.

Let n have base-p digits $n_0, \ldots, n_L$ with $n_L \geq 1$, and let t be the number of digits $p - 1$ at the end of n, so $n_0 = \cdots = n_{t-1} = p - 1$ and either $t = L + 1$ or $n_t \leq p - 2$. Then $n + 1 = p^t\left(\lfloor n/p^t \rfloor + 1\right)$, and when $t \leq L$ the last digit of $\lfloor n/p^t \rfloor + 1$ is $n_t + 1 \in [1, p - 1]$. So $m = t$ and $c = \lfloor n/p^t \rfloor + 1$ (and $c = 1$ when $t = L + 1$). By Kummer, it suffices to find the largest number M of carries in $k + l$ over all $k + l = n$ with $k, l \geq 0$; the claim is that $M \leq 1$ if and only if $c < p^2$.

We use the carry equation $k_i + l_i + c_i = n_i + pc_{i+1}$ with $c_0 = 0$. *Upper bound.* If $i < t$ and $c_i = 0$, a carry out of position i needs $k_i + l_i = n_i + p = 2p - 1$, which is impossible as digits are at most $p - 1$. By induction $c_1 = \cdots = c_t = 0$. There is no carry out of position L, since otherwise $n \geq p^{L+1}$. So carries can only leave the positions $t, t + 1, \ldots, L - 1$, and $M \leq \max(L - t, 0)$.

Construction. Suppose $t \leq L - 1$. Take $k_i = p - 1$, $l_i = 0$ for $i < t$ (no carries); $k_t = p - 1$, $l_t = n_t + 1 \leq p - 1$, which gives $k_t + l_t = n_t + p$ and a carry; $k_i = p - 1$, $l_i = n_i$ for $t < i < L$, which with the incoming carry gives $n_i + p$ and another carry; and $k_L = 0$, $l_L = n_L - 1$, which with the incoming carry gives n_L. These digits add up to n with $L - t$ carries. Hence $M = L - t$ when $t \leq L$, and $M = 0$ when $t = L + 1$.

Conclusion. If $t = L + 1$, then $c = 1 < p^2$ and $M = 0$. If $t \leq L$, then $\lfloor n/p^t \rfloor$ has $L - t + 1$ digits. When $L - t \leq 1$ it is less than p^2, so $c \leq p^2$, and $c \neq p^2$ because $p \nmid c$; so $c < p^2$. When $L - t \geq 2$ it is at least p^2, so $c > p^2$. Therefore $M \leq 1$ if and only if $c < p^2$, as required.

For example, with $p = 2$ the rows with no multiple of 4 are those with $n + 1 = 2^m$ or $3 \cdot 2^m$; row 4 fails because $\binom{4}{1} = 4$. This extends the remark after Example 2, which is the same argument with $M = 0$.